

# SUDHARSAN UDHAYAKUMAR

Data Scientist · Data Engineer · AI/ML Engineer · Business Analyst · Software Engineer

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## PROFESSIONAL SUMMARY

Versatile Data & AI Engineer with 1+ year of professional experience across the full spectrum of data and software engineering — building end-to-end ETL/ELT pipelines, ML/DL models, RAG pipelines, agentic AI frameworks, full-stack web applications, and BI dashboards on large-scale datasets (10M–20M+ records). Proficient in Python, SQL (BigQuery, PostgreSQL, SQL Server), PySpark, and cloud platforms (Azure, AWS). Experienced translating business requirements into scalable, production-grade data and software solutions — with strong OOP, DSA fundamentals, clean code practices, and cross-functional stakeholder communication. IIT Madras Certified Data Scientist. Available immediately in Bengaluru.

## TECHNICAL SKILLS

|                               |                                                                                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages &amp; Query:</b> | Python (Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, PyTorch), SQL (CTEs, Window Functions, Joins, DDL, PL/SQL), JavaScript / TypeScript, Java (fundamentals), R (awareness)                          |
| <b>AI/ML &amp; GenAI:</b>     | Classical ML · Deep Learning · NLP · LLMs (GPT-4, LLaMA, Claude) · RAG Pipelines (Chunking, Embeddings, Vector Search — FAISS, pgvector) · LangGraph · LangChain · Prompt Engineering · Agentic Workflows  |
| <b>Data Engineering:</b>      | ETL/ELT Pipeline Design · Batch & Real-time Ingestion · Data Warehouse (Star Schema, Dimensional Modelling) · Data Quality & Governance · Anomaly Detection · Apache Spark · Hadoop/Hive (awareness) · dbt |
| <b>Full-Stack &amp; APIs:</b> | React.js · Angular · Node.js · FastAPI · REST API Design · Authentication (JWT/OAuth) · MongoDB · PostgreSQL · Redis · System Design · OOP · SOLID Principles · DSA                                        |
| <b>Cloud &amp; DevOps:</b>    | Azure (App Services, DevOps CI/CD, Functions, Monitor) · AWS (S3, Glue, Lambda, Redshift, EMR — certified) · GCP BigQuery · Docker · Git & GitHub · MLOps · Agile/Scrum                                    |
| <b>BI &amp; Analytics:</b>    | Power BI (DAX, Power Query) · Tableau · Matplotlib · Seaborn · Statistical Modelling · EDA · Data Profiling · Stakeholder Reporting · PowerPoint · Excel                                                   |

## PROFESSIONAL EXPERIENCE

### Software Engineer Trainee – Data, AI & Full-Stack | KAAR Infotech Pvt. Ltd., Bengaluru

May 2024 – May 2025

*Clients: Microsoft AI Tour 60, Microsoft AI Tour 12, Microsoft Ignite*

- Architected and owned end-to-end ETL/ELT data pipelines on GCP BigQuery and PySpark — ingested 10M+ records from multi-format sources (CSV, SQL, REST APIs) into a 3-layer Data Warehouse (staging → cleansed → analytical); applied dimensional modelling (star schema), BigQuery partitioning & clustering for 40% faster query execution; integrated AWS (S3, Glue, Redshift) for multi-source data ingestion.
- Developed and deployed production ML/DL models — classical ML (regression, classification, anomaly detection — 85% accuracy) and deep learning NLP models (TensorFlow, Scikit-learn); built and shipped RAG pipelines (chunking, embeddings, vector search, reranking, grounded citations) and agentic AI frameworks (LangGraph, LangChain — tool calling, multi-agent orchestration, memory, fallback paths) integrated into enterprise full-stack applications for Microsoft clients.
- Built full-stack web applications and REST APIs — Angular/React frontend (state management, responsive UI), Python/FastAPI & Node.js backend (JWT auth, secure data handling), PostgreSQL + MongoDB database integration; deployed on Azure with Docker CI/CD pipelines; gathered and translated business requirements into technical specifications for cross-functional stakeholders across 60+ sessions — improving delivery alignment by 40%.
- Built Power BI dashboards (12 DAX measures — KPIs, trend analysis, anomaly alerts), Matplotlib/Seaborn visualisations, and structured PowerPoint reports; implemented multi-stage data quality validation (schema checks, null handling, outlier detection, integrity verification) reducing data discrepancies 25%; improved reporting efficiency 70%; reduced data prep time 60% — enabling analytics teams to independently consume curated datasets.

## KEY PROJECTS

### GenAI RAG & Agentic AI Platform

*Python · LangGraph · LangChain · RAG · LLMs · Angular · FastAPI · MongoDB · Azure · Docker*

- Built production LLM/RAG platform end-to-end — RAG pipeline (chunking, embeddings, FAISS vector search, reranking, citations), LangGraph agentic framework (tool calling, multi-agent orchestration, memory, fallback paths), Angular frontend, Python REST API backend, MongoDB; deployed on Azure CI/CD with Docker — enterprise-scale Microsoft client delivery.

### AutoClean — AI-Powered Data Pipeline Tool (Open Source · Enterprise Adopted)

*Python · PySpark · FastAPI · SQL · Streamlit · MongoDB · PostgreSQL · Azure · CI/CD · GitHub*

- Built automated data quality & ETL tool — null imputation, schema validation, outlier detection (IQR/Z-score), duplicate removal, quality scoring on 15M+ records using PySpark; Streamlit frontend, REST API layer, MongoDB/PostgreSQL, Docker, Azure CI/CD; reduced data prep time 60%; adopted as shared enterprise pipeline across client workflows.

## EDUCATION

### B.E. – Electronics & Communication Engineering | IFET College of Engineering

2024 | CGPA: 8.49

## CERTIFICATIONS

- Certified Data Scientist** — IIT Madras & HCL GUVI (2025) | ML · Deep Learning · NLP · Generative AI · PySpark · SQL · Statistics · Cloud
- Machine Learning & AI Services on AWS** — IIT Madras (GUVI) · Generative AI Powered Data Analytics — Tata (GUVI) | Advanced Python — CSC | IoT & ML — NPTEL